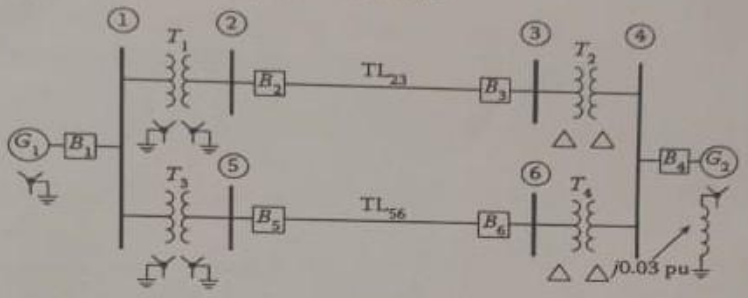
Question1:

a) Determine the symmetrical components for the phase voltages of **(12Marks)**

$Va = 7.3\angle 12.5°, Vb = 0.4\angle -100°, and\ Vc = 4.4\angle 154°V$.

b) Consider the power system shown in next Figure and the given Table below. Assume that each three-phase transformer bank is made of three single-phase transformers. Do the following: (5r

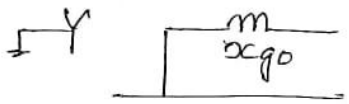
a. Draw the corresponding positive-sequence network.
b. Draw the corresponding negative-sequence network. (2
c. Draw the corresponding zero-sequence network. (2
(3r

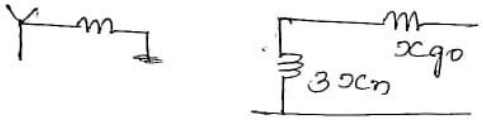


Network Component	MVA Rating	Voltage Rating (kV)	X_1 (pu)	X_2 (pu)	X_0 (pu)
G_1	200	20	0.2	0.14	0.06
G_2	200	13.2	0.2	0.14	0.06
T_1	200	20/230	0.2	0.2	0.2
T_2	200	13.2/230	0.3	0.3	0.3
T_3	200	20/230	0.25	0.25	0.25
T_4	200	13.2/230	0.35	0.35	0.35
TL_{23}	200	230	0.15	0.15	0.3
TL_{56}	200	230	0.22	0.22	0.5

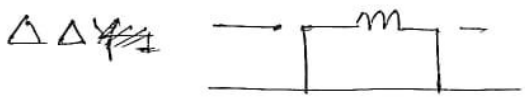
* Zero Sequence network of generator:-

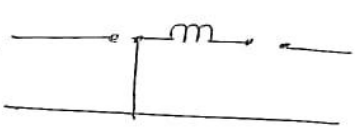
(i) 

(ii) 

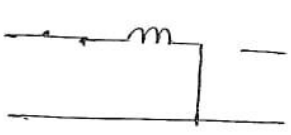
(iii) 

* Zero Sequence network of Transformer:-

(i) 

(ii) 

(iii) 

(iv) 

(a) Symmetrical components of phase voltages

$$\begin{bmatrix} V_{a_0} \\ V_{a_1} \\ V_{a_2} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

where $a = 1\angle 120$ and $a^2 = 1\angle 240$.

Given $V_a = 7.3\angle 12.5^\circ$ V, $V_b = 0.4\angle -100$ & $V_c = 4.4\angle 154$ V

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} V_a + V_b + V_c \\ V_a + (1\angle 120) V_b + (1\angle 240) V_c \\ V_a + (1\angle 240) V_b + (1\angle 120) V_c \end{bmatrix}$$

$$V_{a0} = \frac{1}{3}(7.3\angle 12.5^\circ + 0.4\angle -100^\circ + 4.4\angle 154^\circ) = \frac{4.396\angle 45.1116}{3} \text{ V}.$$

$$\boxed{V_{a0} = 1.465\angle 45.1116^\circ \text{ V}.}$$

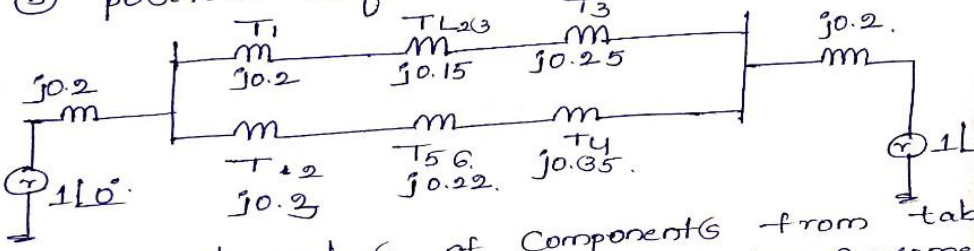
$$V_{a1} = \frac{1}{3}[7.3\angle 125 + (1\angle 120)(0.4\angle -100) + (1\angle 240)(4.4\angle 154)]$$

$$\boxed{V_{a1} = 3.96912\angle 20.5637^\circ \text{ V}.}$$

$$V_{a2} = \frac{1}{3}[7.3\angle 12.5 + (1\angle 240)(0.4\angle -100) + (1\angle 120)(4.4\angle 154)]$$

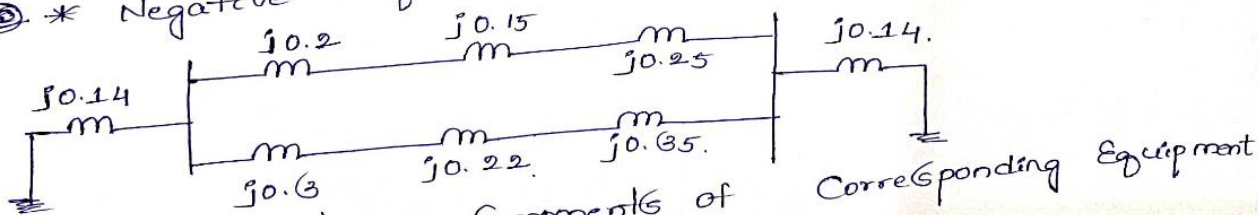
$$\boxed{V_{a2} = 2.52635\angle -19.701^\circ \text{ V}}$$

(b) positive sequence network:-

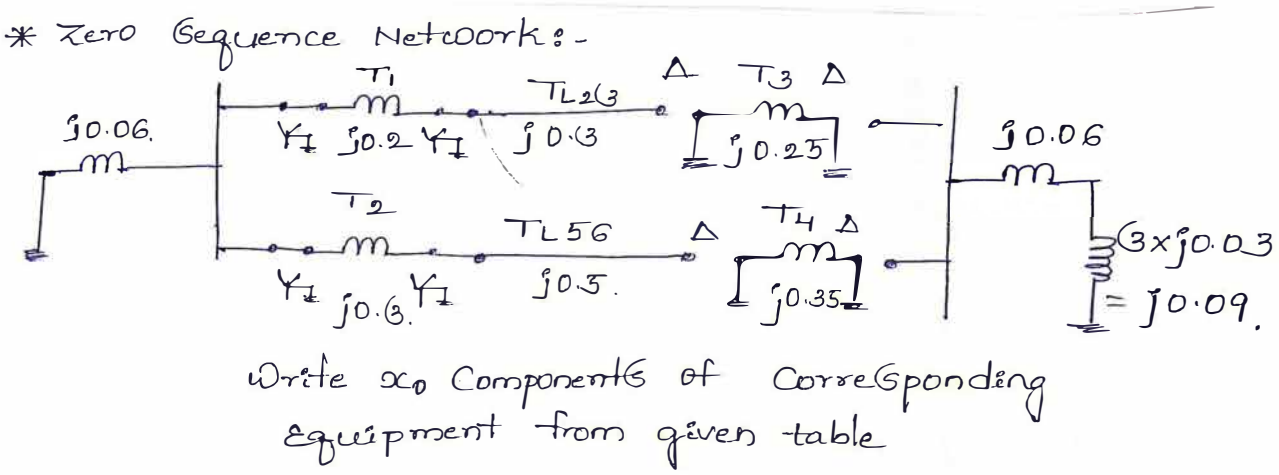


write values of components from table i.e., x_1 components of corresponding equipment.

(c) * Negative sequence network:-



write x_2 components of corresponding equipment from given table.



What is meant by Transposition of the line conductors?

Transposition of line conductors means changing the position of the three phases on the line supports twice over the total length of the line. The line conductors in practice are so transposed that each of the three possible arrangements of conductors exist for one third of the total length of the line.

If the frequency of the transmission line increases then:
a) Line resistance increases
b) Line resistance decreases
c) Shunt rectance increases
d) Shunt reactance decreases

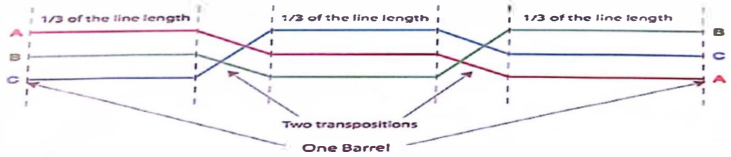
Answer (d). If the frequency of the line increases, then shunt reactance = 1/(2 x pi x f x c). So when frequency (f) inceases shunt reactance decreases.

The phenomenon in which the conductor surface carries more current compared to core when alternative voltage is applied is

Which is the top most conductor in the transmission:
a) R-Phase Conductor
b) S-Phase Conductor
c) T-Phase Conductor
d) Earth Conductor

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If the frequency of the transmission line increases then:
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Answer (d). If the frequency of the line increases, then shunt reactance = 1/(2 x pi x f x c). So when frequency (f) inceases shunt reactance decreases.

The phenomenon in which the conductor surface carries more current compared to core when alternative voltage is applied is
a) Corona
b) Skin Effect
c) Ferranti Effect
d) Lenz's Law

Answer (a). This phenomeon is called Skin Effect

The function of steel wire in the ACSR conductors is to:
a) Compensate for Skin Effect
b) Provide additional mechanical strenght
c) Carry large currents
d) To Reduce Inductance

Answer (b). Galvanised Steel in the ACSR conductors provide better mechanical strenght while the aluminum strands on the outer diameter used to carry the bulk current.

Which material is used for manufacturing of ground wire:
a) Aluminum
b) Steel wire
c) Cast iron
d) Galvanised Steel

Answer (d). Galvanised Steel wire is cheap in cost and withstand corrosion, so this is used as ground wire.

In overhead lines for transmitting power we generally use:
a) Copper Conductors
b) Aluminum Conductors
c) ACSR conductors
d) Galvanized Steel Conductors

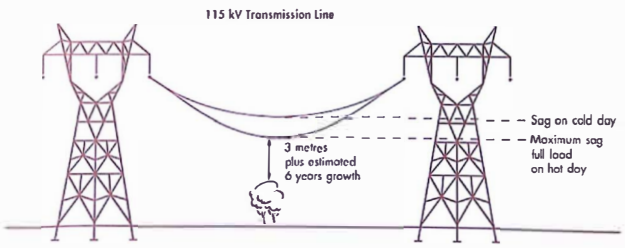
ACSR Conductors implies:
a) Aluminum Conductors Steel Reinforced
b) Anodised Core Steel Reinforced
c) All Conductors Surface Reinforced
d) None of These

ACSR Conductors are used in overhead transmission lines in place of copper because:
a) Economy factor
b) They are light weight
c) All of these
d) High Tensile Strenght

Fault Types

- There are two main types of faults
  - symmetric faults: system remains balanced; these faults are relatively rare, but are the easiest to analyze so we'll consider them first.
  - unsymmetric faults: system is no longer balanced; very common, but more difficult to analyze
- The most common type of fault on a three phase system by far is the single line-to-ground (SLG), followed by the line-to-line faults (LL), double line-to-ground (DLG) faults, and balanced three phase faults

Effect of Temperature rise in overhead lines is to:
a) Increases the SAG and decreases the Tension of lines
b) Decreases the SAG and Increases the Tension of lines
c) Increase in both SAG and Tension of the lines
d) Decreases in both the SAG and Tension of the lines



What is transient state of the power system?

The state of the system in the first second following a system fault or large disturbance is called the transient state of the power system.

What is the need for base values?

The components of various sections of power system may operate at different voltage and power levels. It will be convenient for analysis of power system if the voltage, power, current and impedance ratings of power system components are expressed with reference to a common value called base value. Then all the voltages, power, current and impedance ratings of the components are expressed as a

How transmission voltage is selected

Transmission voltage is stepped up from power plant in order to reduce the losses in transmission. But for every transmission line there is a particular fixed limit for voltage to be transmitted. Beyond this limit the rise in voltage will not result in any economical advantage. Cost analysis is required for finding the optimum voltage; this analysis includes calculating the cost of the conductors, cost of insulators, supports, transformers switchgear, lightning arresters and cost of erection. All the tota cost should be minimum for a particular voltage.

What is meant by symmetrical fault?

The fault is called symmetrical fault if the fault current is equal in all the phases. This fault conditions are analyzed on per phase basis using Thevenin's theorem or using bus impedance matrix. The three-phase fault is the only symmetrical fault.

The real power depends on
a)power factor b)power angle c)power reactive d)voltage

The most common buses in a power system are
a)slack bus b)generator bus c)load bus d)A&B

Why faults occur in a power system?

Faults occur in a power system due to insulation failure of equipment's, flashover of lines initiated by a lightening stroke, permanent damage to conductors and towers or accidental faulty operations.

How are the faults classified? In one method, the faults are classified as,

Symmetrical faults - fault currents are equal in all the phases and can be analyzed on per phase basis

Unsymmetrical faults – fault currents are unbalanced and so they can be analyzed only using symmetrical componenets

List the advantages of per unit computations.

(1) The per unit impedance referred to either side of a single phase transformer is the same
(2) The per unit impedance referred to either side of a single phase transformer is the same.
(3) The per unit impedance referred to either side of a three phase transformer is the same regardless of the three phase connections whether they are Y-Y. Δ-Δ or Δ-Y
(4) The chance of confusion between the line and phase quantities in a three phase balanced system is greatly reduced.
(5) The manufacturers usually provide the impedance values in per unit
(6) The computational effort in power system is very much reduced with the use of per unit quantities.

Which sequence network contains a voltage source
a)Negative b)zero c)Equal d)positive

The reference bus for positive sequence network is system
a)Neutral b)negative c)positive d)zero

The total number of sequence voltage components in a three phase system is
a)10 b)9 c)5 d)6

What is the need for short circuit studies or fault analysis?

The short circuit studies are essential in order to design or develop the protective schemes for various parts of the system. The protective scheme consists of current and voltage sensing devices, protective relays and circuit breakers. The selection of these devices mainly depends on various currents that may flow in the fault conditions.

What are symmetrical components?

An unbalanced system of N related vectors can be resolved into N systems of balanced vectors called symmetrical components.

Positive sequence components
Negative sequence components
Zero sequence components

In which fault zero sequence network is not connected
a)line to ground b)line to line
c)ground to neutral d)line to neutral

What are the factors that affect the transient stability?

The transient stability is generally affected by two factors namely, (1) Type of fault (2) Location of fault.

What is meant by stability study?

The procedure of determining the stability of a system upon occurrence of a disturbance followed by various switch off and switch on actions is called a stability study.

What is meant by short circuit fault?

Short circuit faults involve power conductor or conductors-to-ground or short circuit between conductors. These faults are characterized by increase in current and fall in voltage and frequency.

Bundled conductors used in transmission to reduces the
a) Surge impedance of line
b) Voltage drop of the line
c) corona loss of the line
d) I^2R loss of the line



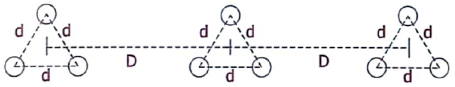
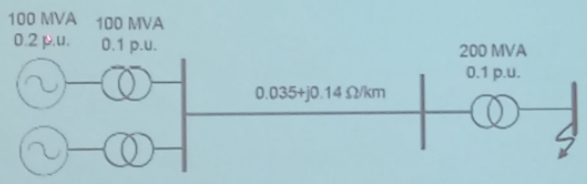


Fig-E: GMD of Bundled conductors. The bundled conductors are placed horizontally. The distance between two extreme conductors is '2D'

Two 100 MVA, 20 kV turbo generators (each of transient reactance 0.2 p.u.) are connected, each through its own 100 MVA, 0.1 p.u. reactance transformer, to a common 132kV busbar. From this bus bar, a 132kV feeder, 40 km in length, supplies an 11 kV load through a 132/11 kV transformer of 200 MVA rating and reactance 0.1 p.u. If a balanced three-phase short circuit occurs on the low-voltage terminals of the load transformer, determine, using a 100 MVA base, the fault current in the feeder and the rating of a suitable circuit breaker at the load end of the feeder. The feeder impedance per phase is (0.035 +j0.14) Ω/km.



On 100 MVA base

$$Z_{base} = \frac{132^2}{100} = 174.24\Omega$$

Line impedance in p.u. = 0.008 + j0.032 p.u.

$$Z_{eq} = \frac{0.2+0.1}{2} + 0.032 + 0.1 \times \frac{100}{200} = 0.232 \text{ p.u.}$$

$$\text{Fault level} = \frac{100}{0.232} = 431 \text{ MVA}$$

A generator rated at 400 MW, 0.8 power factor, 20 kV has a star-connected stator winding which is earthed at its star point through a resistor of 1 Ω. The generator reactances, in per unit on rating, are:
$X_1 = 0.2$, $X_2 = 0.16$, $X_0 = 0.14$
The generator feeds a delta-star-connected generator transformer rated at 550 MVA which steps the voltage up to a 275 kV busbar. The transformer star-point is solidly earthed and the transformer reactance is 0.15 p.u. on its rating. The 275 kV busbar is connected only to the transformer. Assume that for the transformer $X_1 = X_2 = X_0$.
Using a base of 500 MVA calculate the base current and impedance of each voltage level.

Calculate the fault current in amperes for:
(a) a 275kV busbar three-line fault;
(b) a 275 kV single-line-to earth fault on the busbar;
(c) a 20 kV line -line fault on the generator terminals;
(d) a 20 kV line-to-line-to-earth fault on the generator terminals.

On 500 MVA, 275 kV base

$I_{base} = 500 \times 10^6 / (\sqrt{3} \times 20 \times 10^3) = 1049.7 A$

$Z_{base} = 20^2 / 500 = 0.8\ \Omega$

Generator MVA = 400/0.8 = 500 MVA

For power-system fault studies, it is assumed that the system is operating under balanced steady-state conditions prior to the fault, and sequence networks are uncoupled before the fault occurs.
(a) True (b) False

The first step in power-system fault calculations is to develop sequence networks based on the single-line diagram of the system, and then reduce them to their Thevenin equivalents, as viewed from the fault location.
(a) True (b) False

When calculating symmetrical three-phase fault currents, only _____ sequence network needs to be considered.

Positive

In order of frequency of occurance of short-circuit faults in three-phase power systems, list those: _____, _____, _____, _____.

1. Line-to-Ground (L-G)
2. Line-to-Line (L-L)
3. Double Line-to-Ground (LL-G)
4. Three-Phase (L-L-L)

[6] Power transformers are generally designed to have maximum efficiency around
A. No load
B. Half load
C. Near full load
D. 10% overload
[7] The transformer laminations are insulated from each other by
A. Mica strip
B. Thin coat of varnish
C. Paper
D. Any of the above

[8] A transformer core is laminated to
A. Reduce hysteresis loss
B. Reduce eddy current losses
C. Reduce copper losses
D. Reduce all above losses